

Things You Wanted to Know!

Q1. Why is the receiver of a land-line telephone connected to the dialer using a spiral wire? Does it have any specific significance? Why not use a simple wire?

Pooja Math
Kalaburagi

A1. There are two advantages of using a coiled wire. First, straight cords tend to be long. A coiled cord typically offers more flexibility as it can extend to allow for a larger reach. It can stretch to almost double of its un-stretched length and snap back to its original size, time after time. Second, it also prevents tangling.

There are some downsides also. It can be a safety hazard if not used properly. If stretched too far and released, it can spring back with force, which can cause damage to things or injury to someone.

Q2. I am a structural engineer and I need to install an automatic remote control system, alarm system and a CCTV at my home. What will I need for installing these systems? Will I need to install a separate wire-line? Please help me.

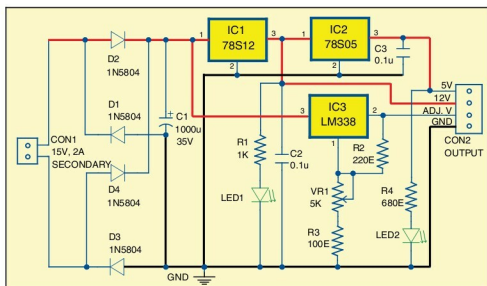
Venkat Raj
Salem

A2. This is a specialised field. There is no off-the-shelf solution available for this. You should consult with companies that provide security services.

Q3. I am working on a project of variable voltage and variable current power supply (DC). Can you help me with this project?

Dinesh Variyani
Through e-mail

A3. Designing a power supply is based on voltage and current requirements of the project. You may have to consider additional require-



Circuit diagram of a power supply

ments, such as protection against short circuit, over-loading and metering facility. Selection of components, such as diodes for rectification and voltage regulators, also depends on voltage and current requirements.

The circuit diagram of a power supply is given in the figure. It provides adjustable voltage up to 12V and fixed voltages of 12V and 5V. Line and load regulation is provided by using conventional series regulators. The current capacity of the power supply is 2A.

The mains AC voltage is stepped down to 15V with the help of a step-down transformer (not shown). The stepped-down voltage is applied to a bridge rectifier circuit formed by 1N5804 diodes, D1 through D4. The rectified voltage is smoothened by capacitor C1, applied to series regulator 78S12 and adjustable regulator LM338. Output of 78S12 regulator is 12V regulated and is available for use at connector CON2. This output is also applied to series regulator 78S05, which provides 5V regulated for powering TTL series of ICs and microcontrollers.

A better arrangement would be the use of a dual secondary-core transformer with lower voltage at the second secondary and separate bridge rectifier circuit to feed an LM7805. Series regulators have the problem of power (as heat) dissipation from the

regulator, which is directly proportional to the voltage difference across it and current taken from the regulator. To minimise power dissipation, we have to minimise the voltage difference across regulators. Proper heat sinks must be used with the regulators.

LM338 is an adjustable three-terminal positive-voltage regulator, capable of supplying in excess of 5A over a 1.2V to 32V output range. Under sustained loading conditions, the current limit decreases to a safe value, protecting the regulator. Thermal overload protection is also available on the chip. Overload protection remains functional even if the adjustment pin is accidentally disconnected. The device provides excellent line regulation. It is easy to use and requires only two resistors, one (R2) between pins 1 and 2, and second between pin 1 and GND. In the circuit, the latter is achieved by resistor R3 + VR1. The output is adjusted using VR1 to the required voltage, which is available at CON2.

LED1 and LED2 are indicators of 12V and 5V, respectively. The common ground of the power supply is at CON2.

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender